CSE350: Digital Electronics & Pulse Techniques

Lecture 04: RTL Circuits: Power Dissipation, Noise Margin & Fanout Calculation

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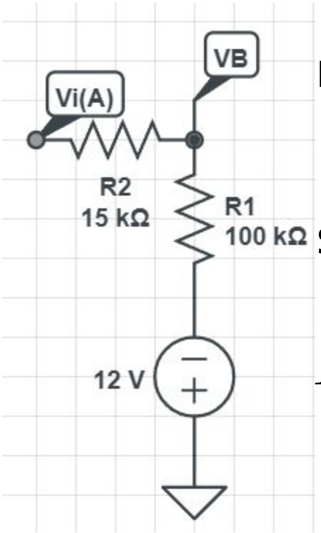
Ex-1: *Find the power dissipated in the RTL inverter circuit.*

Case(0): The BJT is in cutoff for input 0, as we have seen from previous lecture.

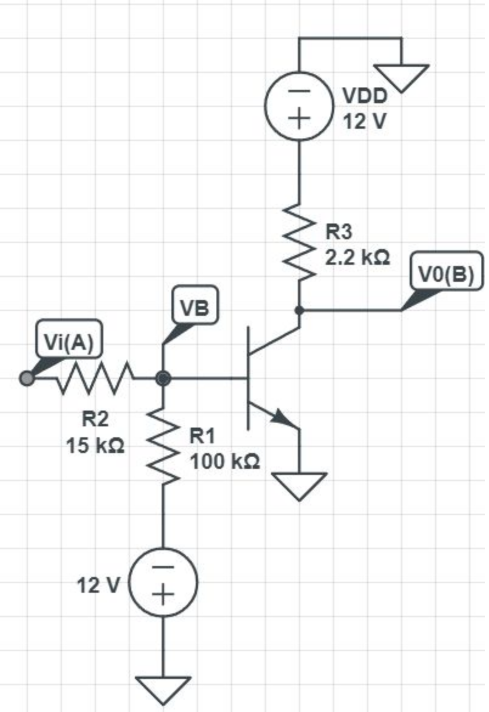


Thus it is replaced by open ckt. The resulting current through R1 & R2,

$$I_1 = \frac{0-(-12)}{115k} = 0.104mA$$

So, dissipated power,

$$P_1 = (0-(-12)) * I_1 = 1.2521mW$$



Case(1): Similarly, from previous lecture, we know that the BJT operates in saturation mode for input 1.

For which we found, $I_C = \frac{12-0.2}{2.2k} = 5.36mA$ $\qquad I_1 = \frac{12-0.8}{15k} = 0.746mA$

$$I_2 = \frac{0.8-(-12)}{100k} = 0.128mA \qquad I_B = I_1 - I_2 = 0.618mA$$

Also, $V_{CE} = 0.2V$ & $V_B = 0.8V$

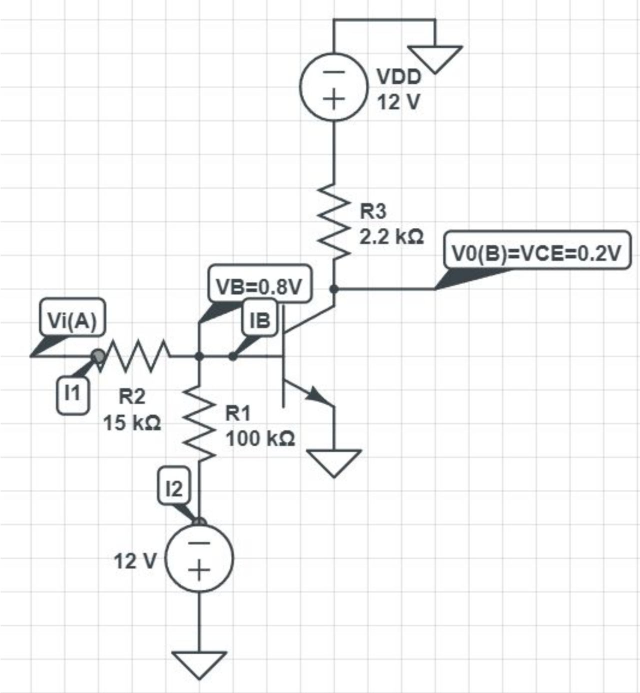
Total power dissipated

$$= (V_i - 0.8) * I_1 + (0.8 - (-12)) * I_2$$

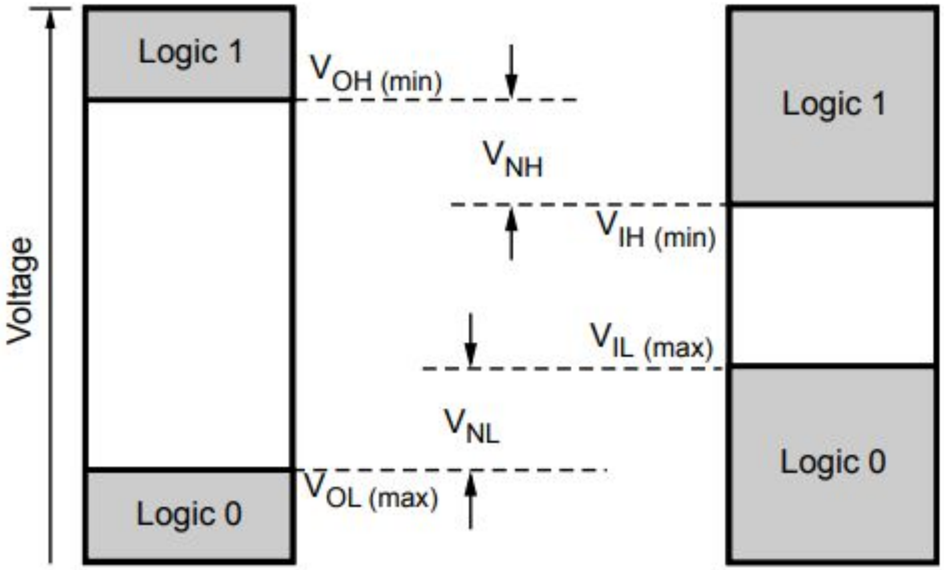
$$+(0.8 - 0) * I_B + (V_{DD} - V_{CE}) * I_C + V_{CE} * I_E$$

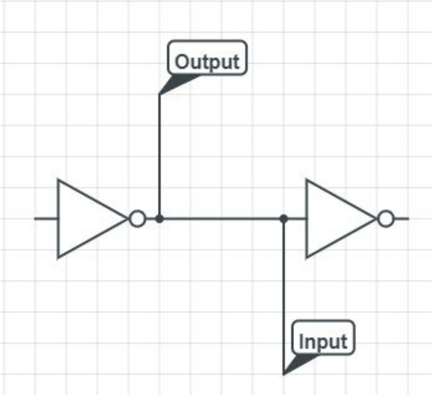
= (12 - 0.8) * 0.746 + (0.8 + 12) * 0.128 + 0.8 * 0.618 + (12 - 0.2) * 5.36

+ 0.2 * 5.36 = 74.8534 mW



Noise Margin: *The maximum amount of noise voltage that can be tolerated by a logic circuit, while completing a successful transmission of signal from output to input.* It occurs due to the unwanted interference of random voltages or signals in transmission line between output of one circuit & input of another consequent one.





Here, for logic '1' at input, a reduced voltage level has to be tolerated due to interference of noise. Similarly, an increased threshold must be treated as logic '0' compared to the threshold at output circuit due to noise. Thus we have two separate noise margins corresponding to two logic states:

High state noise margin, $V_{NH} = V_{OH} - V_{IH}$

Low state noise margin, $V_{NL} = V_{IL} - V_{OL}$

Here,

V_{OH} *is the minimum voltage level at an output in a logical '1' state under defined load conditions.*

V_{OL} *is the maximum voltage level at an output in a logical '0' state under defined load conditions.*

Similar are the definitions of V_{IL} & V_{IH} at input.

Ex-2: Now let us consider an example of determining V_{IL} & V_{IH} from known or given V_{OH} & V_{OL} for the RTL inverter we have studied so far.

The typical output maximum voltage for logic ‘0’ state of saturated logic family circuits is 0.2V.

So, V_{OL} = 0.2V

V_{OH} will be provided directly in the question or there will be sufficient information given to find it. Let’s say it is said that the output may drop 0.5V from maximum voltage. In our example, the maximum voltage for logic ‘1’ state is equal to V_{DD}, which is 12V.

Thus, $V_{OH} = 12 - 0.5 = 11.5V$

Now, for finding V_{IL}, we consider the minimum voltage required between base & emitter to turn on a BJT (i.e. the checking condition of cutoff mode, $V_{BE} < 0.5V$), and perform necessary circuit analysis to obtain the corresponding input voltage.

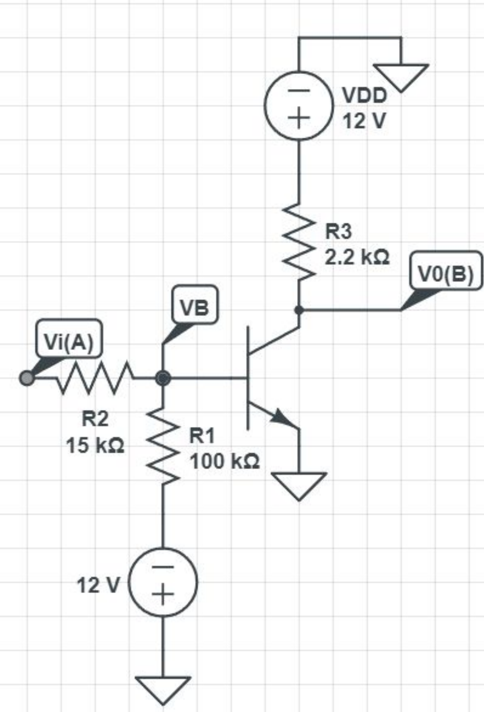
KCL at base,

$$\frac{V_i - V_B}{15k} = \frac{V_B - (-12)}{100k} + I_B$$

But in cutoff, $I_B = 0A$

So, $\frac{V_i - 0.5}{15k} = \frac{0.5 + 12}{100k} + 0$

$\Rightarrow$ $V_i = 2.375V$, this is the *maximum input voltage that can be applied without turning on the BJT*. So, $V_{IL} = 2.375V$



For V_{IH}, we perform a similar analysis, considering the <u>boundary condition between saturation & forward active mode</u> of the BJT. The checking condition in saturation mode was, $\frac{I_C}{I_B} = \beta_{forced} < \beta_{F(Forward)}$. So, the boundary condition will be, $\beta_{forced} \simeq \beta_F$

Again, we use our previous findings for the RTL inverter operating in saturation mode,

$$I_C = \frac{12 - 0.2}{2.2k} = 5.36mA \quad V_{CE} = 0.2V \quad V_B = 0.8V$$

Let's say the forward beta, β_F is 30.

So, $I_B = \dfrac{I_C}{\beta_{forced}} = \dfrac{I_C}{\beta_F} = \dfrac{5.36}{30} = 0.1788mA$

Using this, we can find V_{IH} by applying KCL at base. It gives,

$$\frac{V_{IH} - V_B}{15k} = \frac{V_B - (-12)}{100k} + I_B$$

$$\frac{V_{IH} - 0.8}{15k} = \frac{0.8 + 12}{100k} + 0.1788$$

$$V_{IH} = 5.4018V$$

Note: The region between V_{OL} & V_{OH} is called Abnormal Operation Region and the region between V_{IL} & V_{IH} is known as Intermediate Region.

Now let us calculate the noise margins. We get,

$$V_{HL} = V_{OH} - V_{IH} = 11.5 - 5.4018 = 6.0982V$$

$$V_{NL} = V_{IL} - V_{OL} = 2.375 - 0.2 = 2.175V$$

Well, we got two separate noise margins. Which one should we consider for our logic circuit? It is obvious that since high noise margin is greater than low noise margin, it can easily tolerate noise up to V_{NL}. That cannot be said the other way around. So, we must choose the minimum of the two.

So, *Noise margin = min(6.0982, 2.175) = 2.175V*

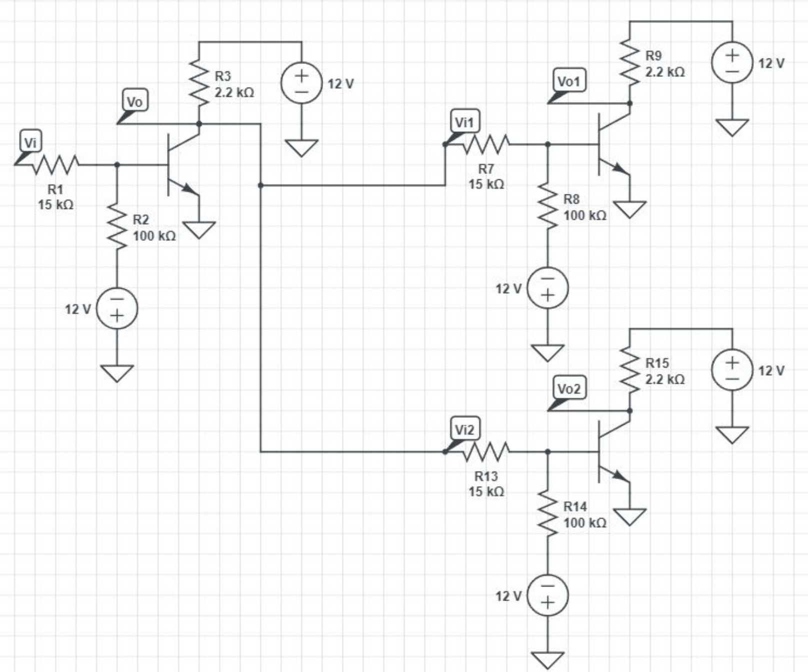
This much noise voltage can be tolerated for both input and output voltages. Noise margin is important when it comes to *comparison among various logic families*. Those with a higher noise margin can tolerate more noise.

Fanout: *The number of input terminals connected to an output terminal is called fanout.* The maximum number of such inputs that can be connected is also the maximum fanout. Usually, when it is asked to find the fanout in a question, it means to determine the maximum fanout, if not stated otherwise. However, if fanout is given as a data of the question, it may or may not be the maximum fanout.

The circuit whose output terminal is being considered in defining a fanout, is called the *driver circuit*. Whereas, the inputs connected to the driver circuit are called *loads*.

The leftmost RTL inverter in the shown circuit is the driver, while

the two others are the loads.



Ex-3: *It is given for the circuit in previous slide that* V_{OH} *= 10V. Calculate the maximum fanout of the circuit.*

To solve this, we need to find the following two for each logic state of the driver circuit:

1. Individual demand of each load
2. Maximum suppliable current from driver

Driver Logic '0': BJT is in saturation. We found that for logic '0', V_{OL} = 0.2V.

So, maximum suppliable current from driver ,

$$I_C = \frac{12 - 0.2}{2.2k} = 5.36mA$$

Now, when the output of the driver is low, consequently, the inputs of the loads will also be low. So, the load inverters' BJTs will be in cutoff. Thus, all their base currents will be 0. So, the currents flow from output of driver to the -12V of the loads. Consequently, current demanded by each load =

$$\frac{0.2 - (-12)}{15k + 100k} = \frac{12.2}{115k} = 0.106mA$$

Thus, maximum fanout = $\left\lfloor \frac{5.36}{0.106} \right\rfloor = \lfloor 50.6 \rfloor = 50$

So, at best 50 inputs can be connected in logic '0' state of the driver.

Driver Logic '1': BJT is in cutoff state & V_{OH} = 10V

Thus, maximum suppliable current = $\frac{12-10}{2.2k} = 0.909mA$

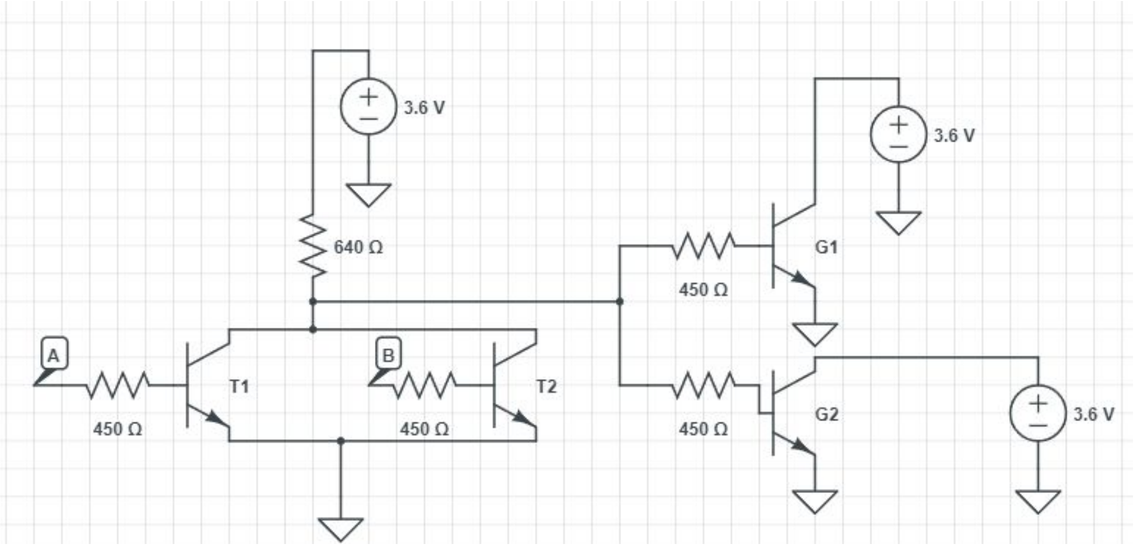
Since driver BJT is off, the full current from the biasing 12V will pass to the loads

Now, the load circuits' BJTs will be in saturation due to high output from driver. So, individual demand of each load = $\frac{10-V_B}{15k} = \frac{10-0.8}{15k} = 0.6133mA$

So, maximum fanout for logic '1' of driver = $\left\lfloor \frac{0.909}{0.6133} \right\rfloor = \lfloor 1.48 \rfloor = 1$

The overall maximum fanout has to be decided based on the worst case scenario, which will be the minimum of the two cases, to avoid malfunctioning of the driver circuit. So, maximum overall fanout = min(1,50) = 1

Practice Problem: *If the number of fanout is 5, calculate the minimum value of β_F, so that these circuits operate as a NOR gate. Assume all transistor have same value of β.*



Driver (NOR)

Load (Inverter)